

Question 1 : Answer the following :

(5 M)

a) When solving a linear programming problem graphically, there are special cases that we encounter, mention them?

1. Alternative optimal solutions
2. Unbounded solution
3. No feasible solution

b) To start the simplex algorithm, there are two requirements that must be met. What are they?

They are:

1. The coefficients of the basic variables (slack variables) form an **Identity matrix**.
2. In the objective function row, the coefficients of the basic variables are **zero**.

Question 2: Solve the following problem Graphically:

(7 M)

$$\text{Maximum } z = 3x_1 + 2x_2$$

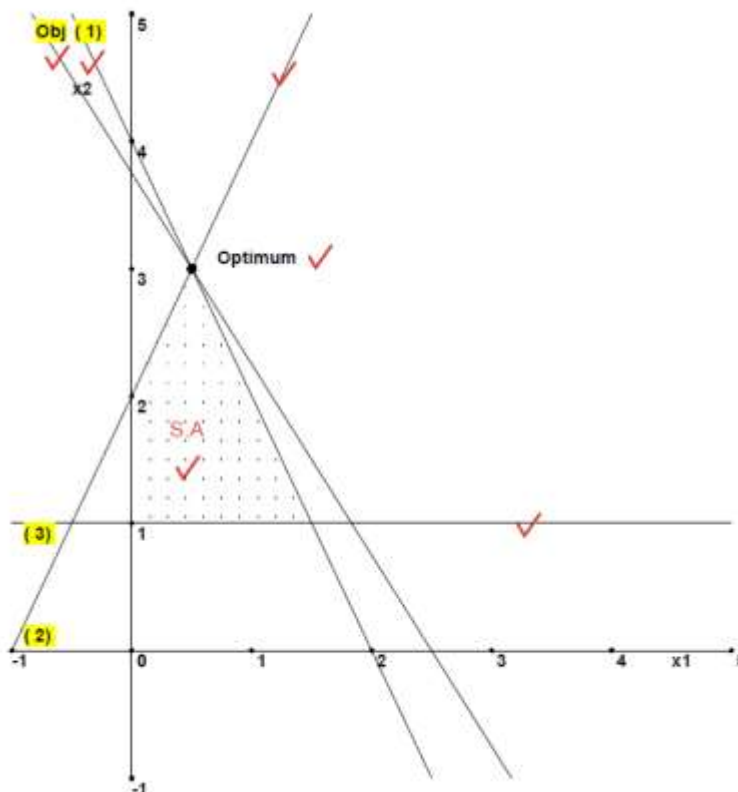
S. to:

$$2x_1 + x_2 \leq 4$$

$$-2x_1 + x_2 \leq 2$$

$$x_2 \geq 1$$

$$x_1 \geq 0, x_2 \geq 0$$



$$Z = 7.5 \text{ at } x_1 = 0.5 \text{ and } x_2 = 3$$

Question 3 : Given the following problem:**(8 M)**

$$\text{Maximum } Z = 8x_1 - 2x_2$$

$$\text{S.to} \quad 2x_1 + x_2 \leq 12$$

$$x_1 - 5x_2 \geq 6$$

$$4x_2 = -7$$

$$x_1 \geq 0, x_2 \text{ Unrestricted in sign.}$$

1. What methods can be applied to solve the given linear programming problem?
2. Write the problem in standard form (equation form). **"Don't solve the problem"**

Solution:

1. Graphical method and Algebraic method. **2**
- 2.

$$\text{Maximum } Z = 8x_1 - 2x_2^+ + 2x_2^- + 0s_1 + 0s_2 \quad \mathbf{2}$$

S.to:

$$2x_1 + x_2^+ - x_2^- + s_1 = 12 \quad \mathbf{1}$$

$$x_1 - 5x_2^+ + 5x_2^- - s_2 = 6 \quad \mathbf{1}$$

$$-4x_2^+ + 4x_2^- = 7 \quad \mathbf{1}$$

$$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0 \quad \mathbf{1}$$

Question 4 : Given the following problem:**(10 M)**

$$\text{Min } Z = x_1 - 2x_2$$

$$\text{S.to} \quad 2x_1 + 3x_3 \leq 1$$

$$3x_1 + 2x_2 - x_3 \leq 5$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

- a) Write the problem in standard form (equation form).
- b) Solve the LP problem using the Simplex method.

Solution:

$$\text{a) Minimize } Z = x_1 - 2x_2 + 0s_1 + 0s_2 \quad \mathbf{1/2}$$

S.to:

$$2x_1 + 3x_3 + s_1 = 1 \quad \mathbf{1/2}$$

$$3x_1 + 2x_2 - x_3 + s_2 = 5 \quad \mathbf{1/2}$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0 \quad \mathbf{1/2}$$

b)

	x1	x2	x3	S1	S2	Sol.
Z	- 1	2	0	0	0	0
S1	2	0	3	1	0	1
S2	3	(2)	-1	0	1	5
Z	- 4	0	1	0	-1	-5
S1	2	0	(3)	1	0	1
x2	1.5	1	-0.5	0	0.5	2.5
Z	- 4.67	0	0	-0.33	-1	-5.33
x3	0.67	0	1	0.33	0	0.33
x2	1.83	1	0	0.17	0.5	2.67

First table 2m (every row 0.5m + 0.5m for pivot element)

Second and third table 3m (1m for every row) , Every mistake in calculation – 0.5m)